Exploring the risk factors associated with Dementia onset: A statistical analysis of health, lifestyle, and genetic factors.

INFO-B518-Applied Statistical Methods for Biomedical Informatics

GROUP 8:

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ABSTRACT

Dementia is a growing public health concern with significant implications for individuals and healthcare systems. This project explores key predictors of dementia onset using statistical methods to analyze health, lifestyle, and genetic factors. A publicly available Kaggle dataset with 1,000 patient records was utilized, including variables such as cognitive test scores, APOE_ϵ4 genetic marker, smoking status, depression, age, and family history.

Exploratory Data Analysis (EDA) provided initial insights through summary statistics and visualizations. Statistical methods included ANOVA for numerical variables (e.g., age, cognitive test scores) and Chi-Square tests for categorical variables (e.g., APOE_ϵ4, education, and smoking status). Predictive modeling with Logistic Regression, Random Forest, and Decision Tree algorithms identified significant predictors and improved accuracy.

Key findings highlighted cognitive test scores, depression status, and smoking history as critical predictors of dementia. Scores below 7.5 strongly associated with dementia, while depression and smoking increased risk. The study underscores the importance of health and lifestyle factors in dementia risk assessment and calls for further research with larger datasets to enhance predictive accuracy.

Keywords: Dementia, Cognitive Test Scores, APOE ε4, Risk Factors, Exploratory Data Analysis, Predictive Modeling, Logistic Regression, Random Forest, Decision Tree, Depression, Categorical Variables, Numerical Variables, Data Preprocessing, ANOVA, Chi-Square Test.

INTRODUCTION

Research Question:

Which health, lifestyle, and genetic factors significantly influence the onset of dementia?

This study explores how variables such as cognitive test scores, APOE ε4 genetic marker, smoking status, depression, and other demographic and health-related attributes contribute to dementia risk.

Background and Significance:

Dementia is a progressive neurodegenerative disorder that affects memory, cognition, and daily functioning, posing a significant burden on individuals, families, and healthcare systems worldwide. While aging is a primary risk factor, other determinants such as genetics, lifestyle, and health conditions also contribute to its onset and progression. Understanding these factors is crucial for early identification, prevention, and intervention to delay or mitigate its impact. This analysis focuses on identifying significant predictors of dementia onset by examining health, lifestyle, and demographic factors in a dataset of 1,000 patients (Adeyemi, 2024). Early risk assessment is essential for enabling timely interventions, personalized care plans, and efficient resource allocation. Data-driven approaches can uncover patterns and relationships among risk factors, providing valuable insights to inform public health policies and clinical decisions (Livingston et al., 2020).

Dataset and Scope:

The study uses a Kaggle dataset of 1,000 patients, analyzing demographic, health-related, and lifestyle factors. Through data preprocessing, statistical tests (ANOVA, Chi-Square), and predictive models (Logistic Regression, Random Forest, Decision Tree), it identifies significant predictors of dementia to support targeted prevention strategies.

DATA DESCRIPTION

Source and Size:

The dataset used in this study is publicly available on Kaggle, uploaded by Timothy Adeyemi in 2024. It contains 1,000 patient records, offering a diverse range of demographic, health-related, lifestyle, and genetic variables.

Variables:

1. **Dependent Variable**:

   Dementia: Binary variable indicating the presence (1) or absence (0) of dementia.
2. **Independent Variables**: Age, Gender, Education Level, Family History, Cognitive Test Scores, APOE ε4 genetic marker, Blood Oxygen Level, Body Temperature, Depression Status, Smoking Status, Physical Activity, Nutrition Diet, Prescription, Dosage, and Medical History.

Variable Types:

Numerical (Continuous): Age, Cognitive Test Scores, Blood Oxygen Level.

Categorical:

Binary: Gender, APOE ε4, Depression Status.

Nominal: Education Level, Smoking Status, Physical Activity.

Preprocessing Steps:

To ensure data quality, missing data in "Prescription" and "dosage in mg" columns was handled by dropping them due to high missing values. Outliers were removed using boxplots and summary statistics. Correlation analysis flagged multicollinearity among numerical variables, and irrelevant columns were eliminated. These preprocessing steps refined the dataset for reliable statistical and predictive modeling.

Descriptive Statistics:

Summary Statistics:

The dataset included both continuous (age, cognitive test scores) and categorical (gender, smoking status) variables. Age had a mean of 70 years, and cognitive test scores showed a median of 7.5, with lower scores linked to dementia. Gender was balanced, and smoking status was categorized for analyzing its impact on dementia onset, offering key insights into risk factors.

Visualizations:

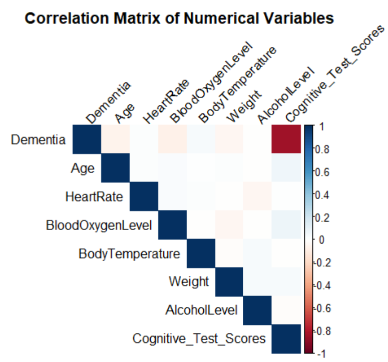

Figure 1. Heatmap: Correlation Analysis of Dementia and Related Variables

The heatmap shows a strong negative correlation (-0.84) between Cognitive Test Scores and Dementia, while blood oxygen level and age have weaker correlations with dementia**.**

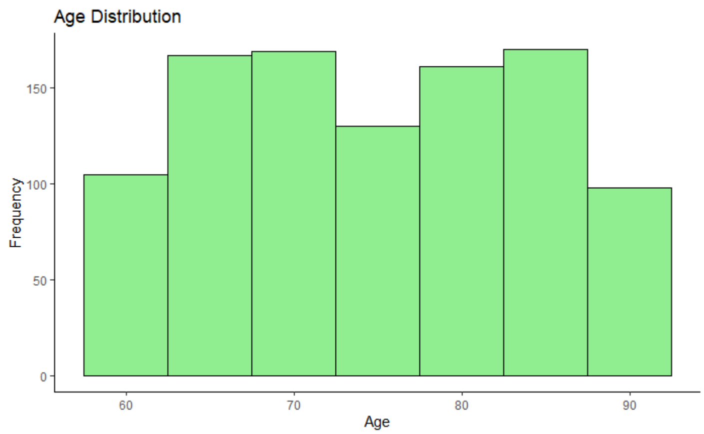

Figure 2. Histograms: Age distribution showed uniformity with peaks in the 60–70 and 80–90 ranges.

The distribution is roughly uniform with no significant skewness. The majority of people are between 60 and 90 years old, with the highest frequencies in the 60-70 and 80-90 age groups.

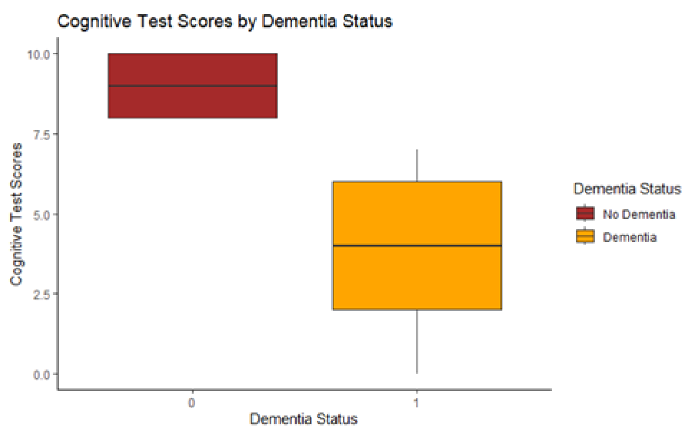

Figure 3. Boxplots: Highlighted significant differences in cognitive test scores between dementia and non-dementia groups.

Dementia group: Median score below 5, with high variability (IQR large).Non-dementia group: Scores significantly higher, with a narrower IQR.

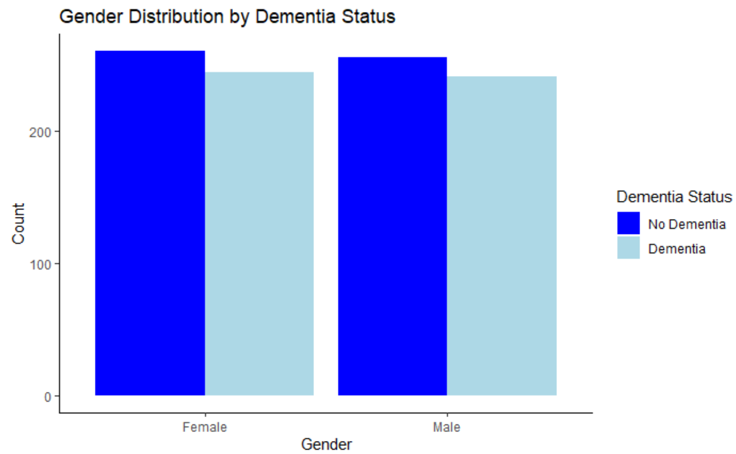

Figure 4. Bar Charts: Gender distribution showed no significant differences in dementia prevalence.

Both genders have similar counts, with slightly higher numbers in the No Dementia category for both males and females.

STATISTICAL METHODS

Hypothesis testing was conducted to assess factors influencing dementia onset. ANOVA identified cognitive test scores as the most significant numerical predictor, while the Chi-Square Test showed significant associations with APOE ε4, family history, and depression. Predictive models, including Logistic Regression, Random Forest, and Decision Tree, highlighted cognitive test scores, depression, and smoking status as key predictors. Logistic Regression faced multicollinearity, while Random Forest showed potential overfitting. Decision Tree offered interpretability. Model performance was validated using accuracy, sensitivity, and AUC metrics, providing robust insights into dementia predictors.

Table 1-Key Statistical Test Results

Variable	Statistical Test	p-value	Significance	Interpretation
Cognitive Test Scores	ANOVA	< 0.001	Significant	Scores below 7.5 strongly associated with dementia.
Age	ANOVA	< 0.05	Significant	Older age is associated with higher dementia risk.
Smoking Status	Chi-Square	< 0.05	Significant	Former smokers show increased risk; current smokers show no significant relationship.
Depression Status	Chi-Square	< 0.001	Significant	Depression strongly associated with dementia presence.
APOE ε4	Chi-Square	< 0.05	Significant	Presence of the APOE ε4 genetic marker increases dementia risk.
Blood Oxygen Level	ANOVA	< 0.05	Significant	Lower levels associated with dementia, though weaker than cognitive scores or depression.

Summarize statistical test-

To evaluate the factors influencing dementia onset, various statistical tests and predictive modeling techniques were applied to the dataset, including ANOVA, Chi-Square tests, and logistic regression. ANOVA revealed that Blood Oxygen Level, Age, and Cognitive Test Scores are significant continuous predictors of dementia, while Chi-Square tests identified significant associations with Education Level, Smoking Status, APOE_e4 gene, Depression Level, and Family History. A logistic regression model showed no significant contributions from Age and Blood Oxygen Level, with large standard errors indicating potential issues like multicollinearity and overfitting. The model's poor fit and high p-values suggest a need for a revised model with better specification and predictive utility

Table 2-Performance Comparison of Machine Learning Models

Model	Accuracy	Sensitivity	Specificity	AUC	Overfitting risk
Logistic Regression	100%	1.0	1.0	1.0	High (Overfit)
Random Forest	99.3%	0.99	0.99	1.0	High (Overfit)
Decision Tree	97.5%	0.98	0.97	0.99	Moderate

Interpretation of Statistical Results

Significant predictors of dementia include Cognitive Test Scores, Depression Status, Smoking Status, and APOE ε4. Gender and Blood Oxygen Level show weak or no association. Logistic Regression faces multicollinearity issues, while Random Forest risks overfitting due to the small sample size.

DISCUSSION

Interpretation of Results:

The analysis identified cognitive test scores below 7.5 as a strong predictor of dementia, emphasizing the importance of cognitive assessments for early intervention. Depression status also showed a robust relationship with dementia, highlighting the role of mental health management in reducing risk. Smoking history emerged as a critical factor, with former smokers showing a higher likelihood of developing dementia, underscoring the long-term neurological effects of smoking. However, the lack of a meaningful relationship for current smokers requires further investigation. These findings align with existing research, reinforcing the significance of cognitive, mental health, and lifestyle factors in dementia prevention.

Significance of Findings:

The findings have practical implications, including using cognitive test scores as a screening tool for early dementia risk identification and addressing depression through mental health interventions. Smoking cessation campaigns should target former smokers due to their heightened risk. Unexpected outcomes, such as the lack of significant relationships for gender and blood oxygen levels, warrant further exploration to understand potential confounding factors.

Limitations:

Key limitations include overfitting in logistic regression and random forest models, the small dataset size, and the use of cross-sectional data, which limited generalizability and the ability to analyze risk factor progression. APOE ε4 carriers and other subgroups may have been underrepresented, potentially underestimating their significance.

Future Research:

Future studies should use larger, more diverse datasets and longitudinal data to track changes in key risk factors over time. Non-significant variables, like gender and blood oxygen levels, should be reassessed in larger datasets. Refining predictive models with regularization techniques, such as LASSO regression, and exploring advanced ensemble models could improve prediction accuracy and reliability.

CONCLUSION

This study identified cognitive test scores, depression status, and smoking history as key predictors of dementia, with cognitive scores below 7.5 being the strongest indicator. These findings highlight the importance of routine cognitive assessments and lifestyle interventions for early dementia detection.

The research reinforces the significance of data-driven approaches in identifying actionable dementia risk factors. However, model overfitting and the small dataset limit the generalizability of the results, suggesting a need for larger, more diverse datasets and refined models.

Future research should incorporate longitudinal data and advanced modeling techniques to improve the reliability of predictions and further explore dementia progression. These efforts will contribute to targeted prevention strategies and early interventions to reduce the global dementia burden.

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